

3_Inequalities

Math 742

1/16/2026

Motivation

When proving most of the theorems in this class, it is helpful to have some common inequalities at your finger tips. Often, these inequalities can simplify your proofs and make them more eloquent. The 6 inequalities that are indispensable are:

1. Markov's Inequality
2. Chebyshev's Inequality
3. Cauchy-Schwarz (for expectations)
4. Jensen's Inequality
5. Hoeffding's Inequality
6. Mills' Inequality

Markov/Chebyshev

Theorem (Markov). Let $X \geq 0$ a.s. with $\mathbb{E}[X] < \infty$. Then for any $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}$$

Note: a.s stands for “almost surely”. What this means mathematically is, $X \geq 0$ a.s $\iff \mathbb{P}(X \geq 0) = 1$

Proof.

$$\mathbb{E}[X] = \int_0^\infty x dF(x) \geq \int_t^\infty x dF(x) \geq \int_t^\infty t dF(x) = t \mathbb{P}(X \geq t).$$

Dividing by $t > 0$ yields the result. $\square$

Now, apply Markov to the non-negative r.v. $(X - \mu)^2$:

$$\mathbb{P}[|X - \mu| \geq t] = \mathbb{P}[(X - \mu)^2 \geq t^2] \leq \frac{\mathbb{E}[(X - \mu)^2]}{t^2} = \frac{\sigma^2}{t^2}.$$

$\square$

This is known as Chebyshev's interval:

$$\mathbb{P}[|X - \mu| \geq t] \leq \frac{\sigma^2}{t^2}.$$

$\rightarrow$ 90% of weak law proofs use this chain.

Cauchy–Schwarz Inequality (for expectations)

Theorem. For random variables X, Y with finite second moments,

$$|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$$

Note: We will use this when we cover covariance

$$0 \leq \mathbb{E}[(\lambda X + Y)^2] = \lambda^2 \mathbb{E}[X^2] + 2\lambda \mathbb{E}[XY] + \mathbb{E}[Y^2].$$

The quadratic in λ has non-positive discriminant:

$$(2\mathbb{E}[XY])^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0 \Rightarrow |\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

□

Jensen's Inequality

To prove Jensen's inequality, we must use the notion of a convex function.

A function f is *convex* on an interval I if for any 2 points $(x, f(x))$ and $(y, f(y))$, if you connect them with a straight line, the function is *convex* if it lies below the line.

For more than 2 points, convexity states that a *convex* function lies below a straight line interpolation of the points. If $x_1, x_2, \dots, x_n \in I$ and $p_1, p_2, \dots, p_n$ where $\sum_i p_i = 1$, then

$$f\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i f(x_i)$$

Theorem (Jensen). Let f be convex on an interval I . Then for any r.v. X with range in I ,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

So convex functions “pull expectations upward.”

Proof (Discrete Case): Let f be convex on an interval I and let X be a r.v. taking values $x_1, \dots, x_n \in I$ with probabilities $p_i = \mathbb{P}(X = x_i)$ ($p_i \geq 0, \sum_i p_i = 1$).

Then,

$$\mathbb{E}[X] = \sum_{i=1}^n p_i x_i, \quad \mathbb{E}[f(X)] = \sum_{i=1}^n p_i f(x_i).$$

By convexity of f applied to the convex combination $\sum_i p_i x_i$,

$$f(\mathbb{E}[X]) = f\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i f(x_i) = \mathbb{E}[f(X)].$$

If f is strictly convex, equality occurs iff all x_i with $p_i > 0$ are equal, i.e. X is almost surely constant. □

→ we will use this when we get to maximum likelihood estimation when showing that the MLE for the binomial distribution is unique interior maximizer.

Remark (extension to general r.v.'s) The discrete proof already contains the full idea. Since every integrable random variable can be approximated by finite-valued ones, the same inequality holds for all differentiability assumptions.

Using Jensen's Inequality

Example 1:

Since $f(x) = x^2$ is convex, by Jensen's inequality,

$$[\mathbb{E}(X)]^2 \leq \mathbb{E}(X^2) \implies \mu^2 \leq \mathbb{E}(X^2)$$

$$\text{So, } \mathbb{V}(X) = \mathbb{E}(X^2) - \mu^2 \geq 0$$

Example 2:

Since $f(x) = -\log x$ is strictly convex on $(0, \infty)$, by Jensen's inequality,

$$-\log(\mathbb{E}[X]) \leq \mathbb{E}(-\log X) \implies \log(\mathbb{E}[X]) \geq \mathbb{E}(\log X)$$

This can be used to show that the arithmetic mean is $\geq$ to the geometric mean.

$$AM = \frac{1}{n} \sum_{i=1}^n x_i \geq GM = (x_1 x_2 \dots x_n)^{1/n}$$

Hoeffding's Inequality

Theorem (Hoeffding 1963): Let $X_1, \dots, X_n$ be independent with $a_i \leq X_i \leq b_i$ a.s. Then

$$\mathbb{P}\left(\left|\frac{1}{n} \sum X_i - \mathbb{E}\left[\frac{1}{n} \sum X_i\right]\right| \geq \varepsilon\right) \leq 2 \exp\left(-\frac{2n^2 \varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Hoeffding gives exponential tail decay when variables are bounded, and is much sharper than Chebyshev whenever n is large.

Most often, we use this inequality in the context of the Bernoulli distribution.

Theorem (Bernoulli version): Let $X_1, X_2, \dots, X_n$ be an iid sample from Bernoulli(p), so $0 \leq X_i \leq 1$ and $\mathbb{E}(X_i) = p$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample proportion. Then, for any $\varepsilon > 0$,

$$\boxed{\mathbb{P}(|\bar{X}_n - p| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2)}.$$

The one-sided version is also useful,

$$\boxed{\mathbb{P}(\bar{X}_n - p \geq \varepsilon) \leq \exp(-2n\varepsilon^2)}.$$

Summary

These four inequalities (Markov, Chebyshev, Cauchy-Schwarz, Jensen) are used in literally every proof from sufficiency through Cramér-Rao. Master them.

- Jensen's inequality, used to prove that Bernoulli(p) MLE is concave up (p.319 C&B)
- Cauchy/Schwarz, used to prove Cramer/Rao Inequality (p. 335 C&B)
- Chebyshev, used to prove WLLN (p. 232 C&B)

Mill's Inequality (Gaussian tail bound)

For $Z \sim N(0, 1)$ and $t > 0$,

$$\frac{1}{t\sqrt{2\pi}} e^{-t^2/2} \left(1 - \frac{1}{t^2}\right) < \mathbb{P}(Z > t) < \frac{1}{t\sqrt{2\pi}} e^{-t^2/2}$$

The upper bound is the one we quote on exams:

$$\mathbb{P}(|Z| > t) < \frac{2}{t\sqrt{2\pi}} e^{-t^2/2}$$

Example: How rare is $|Z| > 3$?

Method	Bound	Interpretation / Actual Value
Chebyshev	$\mathbb{P}(Z > 3) \leq \frac{1}{9} \approx 0.111$	"1 in 9" — extremely loose
Mills (upper)	$\mathbb{P}(Z > 3) < \frac{2}{3\sqrt{2\pi}} e^{-9/2} \approx 0.00295$	Upper bound $\approx$ "1 in 339"
Actual value		$\mathbb{P}(Z > 3) \approx 0.00270$ ("1 in 370")

Quote on exams: By Mills' inequality, $\mathbb{P}(|Z| > t) < \frac{2}{t\sqrt{2\pi}} e^{-t^2/2}$

Bottom line: Chebyshev is the sledgehammer of tail bounds, it only uses variance and gives polynomial decay. For the normal distribution, the tails decay exponentially, so we can do way better.

The sharpest simple bound is Mills' inequality (you'll see this again when we prove asymptotic normality of MLEs). Let's see how ridiculous Chebyshev is for the standard normal at 3σ : Chebyshev says $\mathbb{P}(|Z| > 3) \leq 1/9 \approx 0.111$ — that's a 1-in-9 chance of being more than 3 standard deviations away. We all know that almost never happens.

Mills gives $\mathbb{P}(|Z| > 3) < 0.00295$ or about 1 in 339, and the true probability is ≈ 0.0027 (1 in 370). That's over 200 times sharper than Chebyshev. **Moral of the story: Never use Chebyshev on normal tails use Mills instead.**

You'll use this bound when we prove the delta method and asymptotic normality of MLEs later in the semester.

Polynomial decay (slow) $\longleftarrow$ Chebyshev $\longleftarrow$ Hoeffding $\longleftarrow$ Mills $\longleftarrow$ Exact Gaussian tails (fast)

This diagram illustrates how different tail bounds vary in strength, moving from very loose, polynomial-decay bounds like Chebyshev to increasingly sharper, exponentially decaying bounds such as Hoeffding and finally Mills for Gaussian tails. It visually emphasizes that as you move to the right in the hierarchy, the tail estimates become dramatically tighter and more informative.

Final Summary

Inequality	Best use case
Markov	Nonnegative r.v., no moment assumptions except $\mathbb{E}(X)$
Chebyshev	Second-moment bounds; distribution-free
Cauchy–Schwarz	Covariance inequalities, bounding $\mathbb{E}[XY]$
Jensen	Convexity arguments in estimation/entropy
Hoeffding	Bounded random variables; concentration of averages

Inequality	Best use case
Mills	Normal distribution tail bounds; asymptotics